Complex Analysis Notes

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1. Holomorphic Functions

We assume that the reader has a working knowledge of elementary properties of complex numbers. We begin by defining the complex analogue of the real derivative.

Definition 1.1. Let $U \subset \mathbb{C}$ be an open set. A function $f : U \to \mathbb{C}$ is called **holomorphic** at a point $z \in U$ if the following limit exists.

$$\lim_{h \to 0} \frac{f(z+h) - f(z)}{h} \tag{1}$$

When the above quotient exists, we denote it by $f'(z)$. We shall write $f(z) = u(z) + iv(z)$ where $u, v : \mathbb{C} \to \mathbb{R}$. Throughout this section, f shall denote a holomorphic function from an open set $U \subset \mathbb{C}$ to $\mathbb{C}$.

We denote by $\Omega \subset \mathbb{C}$ a nonempty open connected set[1], unless specified otherwise. Such a set is called a *region*. We denote the set of holomorphic functions on Ω by $\mathcal{H}(\Omega)$.

We denote the closed disc about z_0 of radius $r > 0$ by $D(z_0, r) = \{z \in \mathbb{C} \mid |z - z_0| \leqslant r\}$ and the punctured disc by $D'(z_0, r) = D(z_0, r) \setminus \{z_0\}$. The open disc is denoted by $B(z_0, r)$.

It should be observed that holomorphic functions enjoy the "usual rules of differentiation". That is, if $f, g \in \mathcal{H}(\Omega)$ then $f + g, f \cdot g \in \mathcal{H}(\Omega)$. In fact, the chain rule also carries over, indeed if $f \in \mathcal{H}(\Omega)$ and $g \in \mathcal{H}(\Omega')$ where $f(\Omega) \subset \Omega'$, then $h = g \circ f \in \mathcal{H}(\Omega)$.

It is clear that if $f'(z)$ exists at a point $z \in U$ then f is continuous at z since

$$\lim_{h \to 0} f(z+h) - f(z) = \lim_{h \to 0} \frac{f(z+h) - f(z)}{h} \cdot h = f'(z) \cdot 0 = 0.$$

1.1. The Cauchy-Riemann Equations

Let f be holomorphic at $z \in U$. Observe that this limit is taken in $\mathbb{C}$ hence it must be the same regardless of the way we approach z. If we approach z along the real axis, i.e. we restrict h to be real then

$$f'(z) = \lim_{\substack{h \to 0 \\ h \in \mathbb{R}}} \frac{f(z+h) - f(z)}{h} = \frac{\partial f}{\partial x} = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}. \tag{2}$$

Similarly if we approach z along the imaginary axis, i.e. we restrict h to be purely complex then

$$f'(z) = \lim_{\substack{h \to 0 \\ h \in \mathbb{R}}} \frac{f(z+ih) - f(z)}{ih} = \frac{1}{i} \frac{\partial f}{\partial y} = \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial x}. \tag{3}$$

[1] It follows that Ω is path connected.

Comparing equations (2) and (3), we get the **Cauchy-Riemann equations**, namely

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \qquad \text{and} \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

Another result relates the derivative to the Jacobian of f. That is,

$$|f'(z)|^2 = \left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial x}\right)^2 = \frac{\partial u}{\partial x}\frac{\partial v}{\partial y} - \frac{\partial u}{\partial y}\frac{\partial v}{\partial x} = J_f(z).$$

This brings us to our first theorem.

Theorem 1.2. *Let $f : \Omega \to \mathbb{C}$. Then $f \in \mathcal{H}(\Omega)$ if and only if $u, v \in C^1$ and satisfy the Cauchy-Riemann equations.*[2]

Proof. The forward implication has been proved in the discussion above. Let $h = h_1 + ih_2$ and fix $z \in \Omega$. Since $u, v \in C^1$, we may write

$$\begin{aligned} u(z+h) - u(z) &= \frac{\partial u}{\partial x}h_1 + \frac{\partial u}{\partial y}h_2 + h\epsilon_1, \\ v(z+h) - v(z) &= \frac{\partial v}{\partial x}h_1 + \frac{\partial v}{\partial y}h_2 + h\epsilon_2 \end{aligned}$$

where $\epsilon_1, \epsilon_2 \to 0$ as $h \to 0$. Using the above two equations and the Cauchy-Riemann equations, we get

$$f(z+h) - f(z) = \left(\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}\right)(h_1 + ih_2) + h(\epsilon_1 + i\epsilon_2).$$

Hence $f'(z)$ exists, and since z was arbitrary, $f \in \mathcal{H}(\Omega)$. $\square$

We define the Wirtinger derivative operator(s) as

$$\frac{\partial}{\partial z} = \frac{1}{2}\left(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\right), \qquad \text{and} \qquad \frac{\partial}{\partial \bar{z}} = \frac{1}{2}\left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right).$$

It follows that when a function f is holomorphic at z then at that point $\frac{df}{dz} = \frac{\partial f}{\partial z}$. Moreover, the Cauchy-Riemann equations are equivalent to $\frac{\partial f}{\partial \bar{z}} = 0$. This suggests that holomorphic functions are "independent" of $\bar{z}$ and functions of z alone.

1.2. Analytic Functions

Analytic functions and holomorphic functions are closely related, in fact we will prove that they are equivalent. Recall that, given a sequence of complex numbers $\{a_n\}$, the series

$$\sum_{n=0}^{\infty} a_n(z - z_0)^n \tag{4}$$

is a **power series** about the center $z_0 \in \mathbb{C}$. We summarize a few facts about power series with the following theorem:

Theorem 1.3. *For all power series $\sum_{n=0}^{\infty} a_n z^n$ there is an $R \geqslant 0$, called the radius of convergence, with the following properties:*

(i) The series converges absolutely for all $|z| < R$. If $0 \leqslant \rho < R$ then the series converges uniformly for $|z| \leqslant \rho$.

(ii) The series diverges for $|z| > R$.

[2] A better converse is the Looman-Menchoff Theorem.

Proof. We shall show that R is given by $1/R = \limsup_n \sqrt[n]{|a_n|}$. Let $|z| < R$ and fix ρ such that $|z| < \rho < R$. Then $1/\rho > 1/R$ and by the definition of limsup there is an n_0 such that $\sqrt[n]{|a_n|} < 1/\rho, |a_n| < 1/\rho^n$ for all $n > n_0$. Hence $|a_n z^n| < (|z|/\rho)^n$ for all $n > n_0$. We have bounded all but finitely many terms by the geometric series, hence the series converges absolutely.

To show the uniform convergence of $|z| \leqslant \rho < R$, fix ρ' such that $\rho < \rho' < R$. Using the same argument as above, but this time with ρ' instead of ρ, we get that there is an n_0' such that $|a_n z^n| < (|z|/\rho')^n \leqslant (\rho/\rho')^n$ for all $n > n_0'$. Using Weierstrass's test, we may conclude that the series converges uniformly for $|z| \leqslant \rho$.

Finally, for $|z| > R$, fix ρ such that $R < \rho < |z|$. Then $1/R > 1/\rho$ and hence there are infinitely many n for which $|a_n| > 1/\rho^n$. Since $\lim_{n\to\infty} |a_n| \neq 0$, the series diverges. □

We shall now prove that analytic functions are holomorphic in their radius of convergence. In fact, the derivative of an analytic function is itself analytic and has the same radius of convergence.

Theorem 1.4. *The series $\sum_{n=0}^{\infty} a_n z^n$ is holomorphic in its disc of convergence. Moreover, its complex derivative is given by termwise differentiation and has the same radius of convergence.*

Proof. Let

$$f(z) = \sum_{n=0}^{\infty} a_n z^n = S_k(z) + R_k(z)$$

where

$$S_n(z) = \sum_{k=0}^{n-1} a_k z^k \qquad \text{and} \qquad R_k(z) = \sum_{k=n}^{\infty} a_k z^k.$$

Further, let

$$f_1(z) = \sum_{n=1}^{\infty} n a_n z^n.$$

We shall show that $f_1(z) = f'(z)$. For $z \neq z_0$ and $|z|, |z_0| < R$ consider

$$\left|\frac{f(z) - f(z_0)}{z - z_0} - f_1(z_0)\right| \leqslant \left|\frac{S_n(z) - S_n(z_0)}{z - z_0} - S_n'(z_0)\right| + |S_n'(z_0) - f_1(z_0)| + \left|\frac{R_n(z) - R_n(z_0)}{z - z_0}\right|.$$

Let these terms be $A_n(z), B_n$ and $C_n(z)$ respectively. We shall bound each of these terms. Fix $\epsilon > 0$. We may write the $C_n(z)$ as

$$\begin{aligned}
\left|\frac{R_n(z) - R_n(z_0)}{z - z_0}\right| &= \left|\sum_{k=n}^{\infty} a_k \frac{z^k - z_0^k}{z - z_0}\right| \\
&= \left|\sum_{k=n}^{\infty} a_k (z^{k-1} + z^{k-2} z_0 + \cdots + z z_0^{k-2} + z_0^{k-1})\right| \\
&\leqslant \sum_{k=n}^{\infty} k|a_k| R^{k-1}.
\end{aligned}$$

Applying the root test gives us that the above series converges. Hence, there is an n_1 for which $C_n(z) < \epsilon/3$ for all $n > n_1$ and $|z| < R$. Observe that $S_n'(z_0) \to f_1(z_0)$ as $n \to \infty$, so there is n_2 such that $B_n < \epsilon/3$ for all $n > n_2$. Put $n = \max(n_1, n_2)$. By the definition of the derivative, there is $\delta > 0$ such that $0 < |z - z_0| < \delta \implies A_n(z) < \epsilon/3$. This proves $f_1(z) = f'(z)$ for all $|z| < R$.

It is easy to see that the radius of convergence of f' is the same as f since

$$\limsup_n \sqrt[n]{n a_n} = R \cdot \limsup_n \sqrt[n]{n} = R.$$

This follows from the fact that[3] $\lim_{n\to\infty} n^{1/n} = 1$. □

[3] c.f. [Rud76] 3.20.

2. Complex Integration

We define the integration of complex functions of real variables and complex functions of complex variables. The former is relatively easy to do, but for the latter we need to define the notion of curves.

Definition 2.1. A **curve** γ is the continuous function $\gamma : [a, b] \to \mathbb{C}$ where $[a, b] \subset \mathbb{R}$ is an interval. If the initial point $\gamma(a)$ coincides with the final point $\gamma(b)$ then γ is called a **closed curve**. If γ is piecewise continuously differentiable, then it is called a **path**.

It is useful to define some operations on curves. The concatenation of two curves $\gamma_1 : [0, 1] \to \mathbb{C}$ and $\gamma_2 : [0, 1] \to \mathbb{C}$ with $\gamma_1(1) = \gamma_2(0)$, is defined as $\gamma_1 + \gamma_2 : [0, 1] \to \mathbb{C}$ as

$$(\gamma_1 + \gamma_2)(t) = \begin{cases} \gamma_1(2t) & 0 \leqslant t < \frac{1}{2} \\ \gamma_2(2t-1) & \text{otherwise} \end{cases}.$$

The reverse curve of $\gamma : [a, b] \to \mathbb{C}$, is the curve $-\gamma : [a, b] \to \mathbb{C}$ where $t \mapsto \gamma(a + b - t)$.

Let $f : [a, b] \to \mathbb{C}$ be a complex function of one real variable, i.e. $f(t) = u(t) + iv(t)$. We define $\int f = \int u + i \int v$. This satisfies the usual linearity properties. It also satisfies the modulus inequality $|\int f| \leqslant \int |f|$, which we shall now prove.

Let $\theta = \arg \int_a^b f(t)dt$. Now,

$$\left| \int_a^b f(t)dt \right| = \operatorname{Re}\left(e^{-i\theta} \int_a^b f(t)dt \right) = \int_a^b \operatorname{Re}\left(e^{-i\theta} f(t) \right) dt \leqslant \int_a^b |f(t)|dt.$$

We state a theorem from analysis, c.f. [Rud76]. Let $\gamma : [a, b] \to \mathbb{C}$ be a curve. For a partition $P = \{x_0, \ldots, x_n\}$ of $[a, b]$, we define

$$\Lambda(\gamma, P) = \sum_{i=1}^{n} |\gamma(x_i) - \gamma(x_{i-1})| .$$

We define $L(\gamma) = \sup_P \Lambda(\gamma, P)$. If $L(\gamma) < \infty$ then we call γ a rectifiable curve and call $L(\gamma)$ the 'length' of γ. It is a theorem in analysis that

$$L(\gamma) = \int_a^b |\gamma'(t)|dt. \tag{5}$$

Definition 2.2. Let Ω be a region and let $f : \Omega \to \mathbb{C}$. Let $\gamma : [a, b] \to \mathbb{C}$ be a path in Ω. We define

$$\int_\gamma f(z)dz = \int_a^b f(\gamma(t))\gamma'(t)dt.$$

Such an integral is called a *line integral* of f along γ. When integrating along an arc, the positive direction is taken to be anticlockwise by convention.

Observe that a line integral is invariant under the change of parameter. More precisely, if $\alpha : [\alpha, \beta] \to [a, b]$ is an increasing function then the integral of f over $\gamma' = \gamma \circ \alpha$ is the same as over γ. The property follows by simply substituting $u = \alpha(t)$ in the following integral,

$$\int_{\gamma'} f(z)dz = \int_\beta^\alpha f(\gamma(\alpha(t)))\gamma'(\alpha(t))\alpha'(t)dt = \int_a^b f(\gamma(u))\gamma'(u)du = \int_\gamma f(z)dz.$$

The line integral over a closed path γ is invariant under shift of parameter. The old and new initial point determine two arcs γ_1 and γ_2, and the invariance states that the integral over $\gamma_1 + \gamma_2$ is equal to the integral over $\gamma_2 + \gamma_1$.

We now state some properties of line integrals. Let $f : \Omega \to \mathbb{C}$ be continuous and let γ be a path in Ω with parameter interval $[\alpha, \beta]$. if $\gamma = \gamma_1 + \gamma_2$ then

$$\int_\gamma f(z)dz = \int_{\gamma_1} f(z)dz + \int_{\gamma_2} f(z)dz.$$

As a corollary, we have

$$\int_{-\gamma} f(z)dz = -\int_\gamma f(z)dz.$$

An important inequality follows from (5), viz.

$$\left|\int_\gamma f(z)dz\right| \leqslant \int_\alpha^\beta |f(\gamma(t))\gamma'(t)|dt \leqslant L(\gamma) \cdot \sup_{z \in \operatorname{Im}\gamma} |f(z)|.$$

We shall, henceforth, use this result without reference. The following theorem is also sometimes referred to as the *fundamental theorem* for line integrals.

Theorem 2.3. *If f has primitive F on Ω then for any curve γ in Ω, we have*

$$\int_\gamma f(z)dz = f(\gamma(1)) - f(\gamma(0)).$$

Proof. This theorem directly follows from the fundamental theorem of calculus. Indeed,

$$\int_\gamma f(z)dz = \int_0^1 f(\gamma(t))\gamma'(t)dt = \int_0^1 (F(\gamma(t)))'dt = f(\gamma(1)) - f(\gamma(0)).$$

□

This brings us to our first major result.

Theorem 2.4. *(Goursat) Let f be holomorphic on Ω, and let $T \subset \Omega$ be a closed triangle. Then*

$$\int_{\partial T} f(z)dz = 0.$$

Proof. For a closed triangle T, define

$$\eta(T) = \int_{\partial T} f(z)dz.$$

Divide the given triangle T into 4 sub-triangles $T^{(1)}, T^{(2)}, T^{(3)}, T^{(4)}$ by bisecting the sides and joining the points, as shown below.

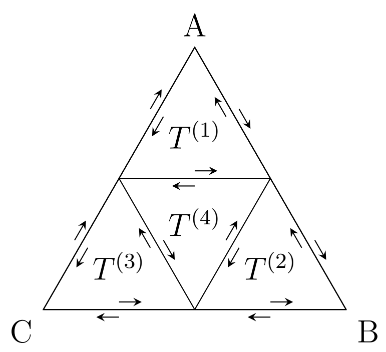


Figure 1: Splitting the integral $\eta(T)$.

Observe that

$$\eta(T) = \eta(T^{(1)}) + \eta(T^{(2)}) + \eta(T^{(3)}) + \eta(T^{(4)}).$$

Hence there is at least one $T^{(k)}$ where $k = 1, 2, 3, 4$; with

$$\eta(T^{(k)}) \geqslant \frac{1}{4}\eta(T). \tag{1}$$

If there are several such subtriangles then they are chosen by some predefined rule which is inconsequential in the proof. Repeating this process, we get a sequence of triangles $\{T_i\}$ such that $T_1 = T$ and $T_i \supset T_{i+1}$ for all $i \in \mathbb{N}$. As a consequence of (1), we have

$$\eta(T_n) \geqslant 4^{-n}\eta(T). \tag{2}$$

Define $d_n = \operatorname{diam} T_n$ and P_n to be the perimeter of T_n for all $n \in \mathbb{N}$. It is easy to see that $d_n = 2^{-n}d$ and $P_n = 2^{-n}P$ where d and P are the diameter and perimeter of T respectively.

Since $\mathbb{C}$ is complete and $\{T_n\}$ is a descending chain of triangles where $d_n \to 0$, it follows from the Cantor Intersection Theorem that $\cap_n T_n = \{z_0\}$. Fix $\epsilon > 0$. Pick $\delta > 0$ such that for $|z - z_0| < \delta$, $f(z)$ is defined and holomorphic and

$$\left| \frac{f(z) - f(z_0)}{z - z_0} - f'(z_0) \right| < \epsilon. \tag{3}$$

Observe that $(z^2/2)' = z$ and $z' = 1$, hence by 2.3, we have $\int_{\partial T_n} dz = 0$ and $\int_{\partial T_n} z dz = 0$. Now,

$$\eta(T_n) = \int_{\partial T_n} (f(z) - f(z_0) - (z - z_0)f'(z_0))dz.$$

Using (3), we get

$$|\eta(T_n)| \leqslant \epsilon \int_{\partial T_n} |z - z_0|\, dz \leqslant \epsilon d_n P_n.$$

Recall that $d_n = 2^{-n}d$ and $P^n = 2^{-n}P$, from (2) we get

$$\eta(T) \leqslant dP\epsilon.$$

Since ϵ was arbitrary, we get that $\eta(T) = 0$. $\square$

Corollary 2.5. *If $p \in \Omega$ is a point such that $f \in \mathcal{H}(\Omega \setminus \{p\})$ and f is defined at p, then the above theorem holds.*

Proof. Suppose p is a vertex A of T, as seen in Figure 1. Pick points R and S on AB and BC respectively such that the perimeter of ARS is less than some ϵ. Then,

$$\eta(T) = \eta(ARS) + \eta(BRS) + \eta(CBS) = \eta(ARS).$$

Since f is bounded on T, our assertion follows. Now for the case where p is any point in T, apply the previous result to ABP, BCP, and CAP. $\square$

It is easily seen that this extends to a finite set of points where f is not holomorphic.

This enables us to prove a particular case of Cauchy's Theorem, it is nonetheless very useful.

Theorem 2.6. *(Cauchy's Theorem for a Convex Set) Let Ω be a convex open set. Let $f \in H(\Omega)$. Then for any closed path γ in Ω, we have*

$$\int_\gamma f(z)dz = 0.$$

Proof. We need only show that f has a primitive on Ω, the theorem then follows from 2.3. For $z_1, z_2 \in \Omega$, let the line connecting them be $\gamma_{z_1,z_2} : [0,1] \to \mathbb{C}$ where $t \mapsto (1-t)z_1 + tz_2$, the image of this path is in Ω since it is convex.

Fix a point $p \in \Omega$. Define $F : \Omega \to \mathbb{C}$ by

$$F(z) = \int_{\gamma_{p,z}} f(w)dw.$$

We now prove that F is the primitive of f on Ω. As a consequence of 2.4, it follows that for any $z' \in \Omega$, we have

$$\int_{\gamma_{p,z}-\gamma_{p,z'}} f(w)dw = \int_{\gamma_{z,z'}} f(w)dw.$$

Now, for $z \in \Omega$,

$$F'(z) = \lim_{h\to 0} \frac{F(z+h)-F(z)}{h} = \lim_{h\to 0} \frac{1}{h} \int_{\gamma_{z,z+h}} f(z)dz = \lim_{h\to 0} \int_0^1 f(z+th)dt$$

We claim that this limit is equal to $f(z)$. Fix $\epsilon > 0$. Since f is holomorphic at z, it is continuous and hence pick $\delta > 0$ such that $|z - z'| < \delta$ implies $|f(z) - f(z')| < \epsilon$. Now for $|h| < \delta$, we have

$$\left| \int_0^1 f(z+th)dt - f(z) \right| = \int_0^1 |f(z+th) - f(z)| dt < \epsilon$$

Hence proved. $\square$

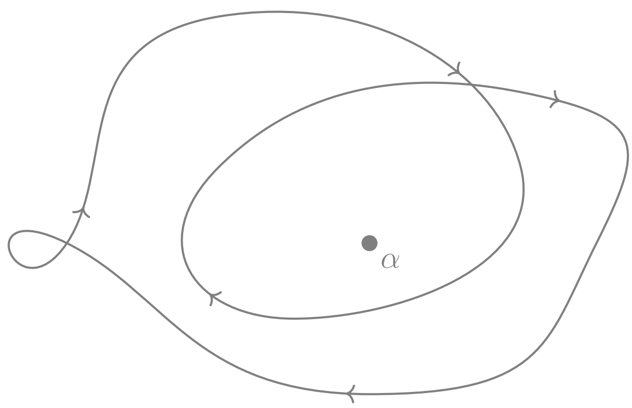
Using 2.5, it follows that the above theorem is valid even if f is not holomorphic at some point $p \in \Omega$!

2.1. Winding Number

Definition 2.7. Let γ be a closed path and $a \in \mathbb{C}$ be any point not in $\operatorname{Im}\gamma$. We define the **winding number** of α with respect to γ as

$$n(\gamma, \alpha) = \frac{1}{2\pi i} \int_\gamma \frac{1}{z-\alpha} dz.$$

The winding number, intuitively, is the number of times a curve γ 'wraps' around a point α. The curve given below has winding number 2.



We now prove some elementary properties of the winding number.

Theorem 2.8. *The winding number $n(\gamma, z)$ is an integer.*

Proof. Let $[\alpha, \beta]$ be the parameter interval of γ and let $S \subset [\alpha, \beta]$ be the (finite) set of points where γ is not differentiable. Then

$$n(\gamma, z) = \frac{1}{2\pi i} \int_\alpha^\beta \frac{\gamma'(s)}{\gamma(s) - z} ds.$$

Define $\varphi : [\alpha, \beta] \to \mathbb{C}$ as

$$\varphi(t) = \exp\left(\int_\alpha^t \frac{\gamma'(s)}{\gamma(s) - z} ds \right). \tag{1}$$

It enough to prove that $\varphi(\beta) = 1$. Differentiating (1), for $t \in [\alpha, \beta] \setminus S$, we get

$$\frac{\varphi'(t)}{\varphi(t)} = \frac{\gamma'(t)}{\gamma(t) - z}.$$

Using this, we can conclude that $\varphi/(\gamma - z)$ is continuous on $[\alpha, \beta]$ whose derivative is zero in $[\alpha, \beta] \setminus S$. It follows that $\varphi/(\gamma - z)$ is constant on $[\alpha, \beta]$. Since $\varphi(\alpha) = 1$, we get

$$\varphi(t) = \frac{\gamma(t) - z}{\gamma(\alpha) - z}$$

Since γ is a closed curve, we have $\gamma(\beta) = \gamma(\alpha)$, whence our result follows. □

Remark 2.9. The winding number is a continuous function in the second slot. That is, for fixed γ, the function $\alpha \mapsto n(\gamma, \alpha)$ is continuous.

Proof. Fix $\alpha_0 \in \mathbb{C} \setminus \operatorname{Im} \gamma$. Further, since $\operatorname{Im} \gamma$ is compact it is closed and hence there is $\delta > 0$ such that

$$\min_t |\alpha_0 - \gamma(t)| = \delta$$

For $\alpha \in B(\alpha_0, \delta/2)$, we have

$$\left| \int_\gamma \frac{dz}{z - \alpha_0} - \int_\gamma \frac{dz}{z - \alpha} \right| = |\alpha - \alpha_0| \left| \int_\gamma \frac{dz}{(z - \alpha)(z - \alpha_0)} \right| \leqslant \frac{2L(\gamma)}{\delta^2} |\alpha - \alpha_0|.$$

Whence, our result follows. □

This shows that γ is integer valued and continuous. Hence $n(\gamma, \alpha)$ is constant on each connected component of $\mathbb{C} \setminus \operatorname{Im} \gamma$. A useful theorem here is that if γ is an injective closed path, i.e. a *Jordan arc*, then it divides the complex plane into exactly two components; one bounded and the other unbounded. This is known as the *Jordan Curve Theorem*.

Further, if S is an unbounded connected component of $\mathbb{C} \setminus \operatorname{Im} \gamma$, then $n(\gamma, \alpha) = 0$ for all $\alpha \in S$. This is clear since we can arbitrarily decrease the value of $1/|z - \alpha|$ for all $z \in \operatorname{Im} \gamma$.

2.2. Cauchy's Theorem(s)

Theorem 2.10. *(Local Cauchy's Integral Formula) Suppose γ is a closed path in a convex open set Ω and $f \in \mathcal{H}(\Omega)$. For $z \in \Omega \setminus \operatorname{Im} \gamma$, we have*

$$f(z) \cdot n(\gamma, z) = \frac{1}{2\pi i} \int_\gamma \frac{f(\zeta)}{\zeta - z} d\zeta.$$

Proof. Fix a point $z \in \Omega$. Define $\varphi : \Omega \to \mathbb{C}$ as the quotient

$$\varphi(\zeta) = \begin{cases} \dfrac{f(\zeta) - f(z)}{\zeta - z} & \zeta \neq z \\ f'(z) & \text{otherwise} \end{cases}$$

Observe that φ is holomorphic at each point of Ω except possibly at z. Recall the corollary to 2.6, it follows that for any closed path γ in Ω

$$0 = \int_\gamma \frac{f(\zeta) - f(z)}{\zeta - z} d\zeta = \int_\gamma \frac{f(\zeta)}{\zeta - z} d\zeta - f(z) \int_\gamma \frac{1}{\zeta - z} d\zeta.$$

Hence proved. $\square$

Theorem 2.11. *(Cauchy's differentiation Formula) Suppose $f \in \mathcal{H}(\Omega)$ where Ω is a convex open set. Let γ be a closed path in Ω and let $z \in \Omega \setminus \operatorname{Im}\gamma$, then*

$$f^{(n)}(z) \cdot n(\gamma, z) = \frac{n!}{2\pi i} \int_\gamma \frac{f(\zeta)}{(\zeta - z)^{n+1}} d\zeta. \qquad (\forall n \in \mathbb{N})$$

Proof. Fix $z_0 \in \Omega$ and a closed path γ in Ω. Since $\operatorname{Im}\gamma$ is compact, it is closed and hence there is $\delta > 0$ such that $|z_0 - \zeta| \geqslant \delta$ for all $\gamma \in \operatorname{Im}\gamma$. Let $0 < r < \delta$. We show that f has a power series expansion about z_0, more precisely, on $D(z_0, r)$. Observe that

$$\frac{1}{\zeta - z} = \frac{1}{(\zeta - z_0) - (z - z_0)} = \frac{1}{\zeta - z_0} \cdot \frac{1}{1 - \dfrac{z - z_0}{\zeta - z_0}} = \frac{1}{\zeta - z_0} \sum_{n=1}^{\infty} \left(\frac{z - z_0}{\zeta - z_0} \right)^n.$$

For $z \in D(z_0, r)$, the last sum converges uniformly since

$$\left| \frac{z - z_0}{\zeta - z_0} \right| \leqslant \frac{s}{r} < 1.$$

For $z \in D(z_0, r)$, upon applying Cauchy's Theorem,

$$\begin{aligned} f(z) \cdot n(\gamma, z) &= \frac{1}{2\pi i} \int_\gamma \frac{f(\zeta)}{\zeta - z} d\zeta \\ &= \frac{1}{2\pi i} \int_\gamma \frac{f(\zeta)}{\zeta - z_0} \sum_{n=0}^{\infty} \left(\frac{z - z_0}{\zeta - z_0} \right)^n d\zeta \\ &= \sum_{n=0}^{\infty} \left(\frac{1}{2\pi i} \int_\gamma \frac{f(\zeta)}{(\zeta - z)^{n+1}} d\zeta \right) \cdot (z - z_0)^n. \end{aligned}$$

We can now use 1.4, hence $n(\gamma, z) \cdot f^{(n)}(z_0) = n! a_n$ where a_n is the n^{th} term in the sum above. $\square$

Corollary 2.12. *(Cauchy's Estimates) Let $f \in \mathcal{H}(\Omega)$ where Ω is open and let $z_0 \in \Omega$. If there is R such that $D(z_0, R) \subset \Omega$ then*

$$|f^{(n)}(z_0)| \leqslant \frac{n!}{R^n} \| f \|_C ,$$

where $\| f \|_C = \sup_{\theta \in [0, 2\pi]} |f(z_0 + Re^{i\theta})|$.

Proof. Let C denote the circle of radius R around z_0. We have

$$|f^{(n)}(z_0)| = \left| \frac{n!}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta \right| \leqslant \frac{n!}{2\pi} \cdot \frac{\| f \|_C}{R^{n+1}} \cdot 2\pi R = \frac{n!}{R^n} \| f \|_C .$$

This proves our assertion. $\square$

Corollary 2.13. *Let Ω be an open set and let $f \in \mathcal{H}(\Omega)$. If $B(z_0, r) \subset \Omega$, then f admits a power series expansion at z_0. That is, for all $z \in B(z_0, r)$, we have*

$$f(z) = \sum_{n=1}^{\infty} a_n (z - z_0)^n \qquad \text{where} \qquad n! a_n = f^{(n)}(z_0).$$

Combining the above corollary and 1.4 gives us that if $f \in \mathcal{H}(\Omega)$ then $f' \in \mathcal{H}(\Omega)$.

Theorem 2.14. *(Morera) Suppose f is a continuous complex valued function in an open set Ω such that*

$$\int_{\partial\Delta} f(z)dz = 0$$

for every closed triangle $\Delta \subset \Omega$, then $f \in \mathcal{H}(\Omega)$.

Proof. Let V be a convex open subset of Ω. Using the same construction as in 2.6, we can construct $F \in \mathcal{H}(V)$ such that $F' = f$. Since the derivatives of holomorphic functions are holomorphic, we get $f \in \mathcal{H}(V)$. Further, since V was arbitrary, we get $f \in \mathcal{H}(\Omega)$. □

A function that is holomorphic in all of $\mathbb{C}$ is called *entire*.

Theorem 2.15. *(Louiville) A bounded entire function is constant.*

Proof. Let f be an entire bounded function. Then there is $M > 0$ such that $|f(z)| < M$ for all $z \in \mathbb{C}$. Using 2.12 with $n = 1$, for any $z \in \mathbb{C}$ we have

$$|f'(z)| < \frac{M}{R}.$$

We can increase the radius R of the circle arbitrarily since f is entire. Hence $|f'(z)| = 0$ for all $z \in \mathbb{C}$. Our result follows. □

Corollary 2.16. *$\mathbb{C}$ is algebraically closed.*

Proof. We induct on the degree of the polynomial. Without loss of generality, assume $P(z) = z^n + a_{n-1}z^{n-1} + \cdots + a_0$. We argue by contradiction, suppose P has no roots in $\mathbb{C}$, then $1/P$ is entire. We show that $1/P$ is bounded, consider

$$\frac{P(z)}{z^n} = 1 + \left(\frac{a_{n-1}}{z} + \frac{a_{n-2}}{z^2} + \cdots + \frac{a_0}{z^n}\right).$$

The terms in the bracket go to 0 as $|z| \to \infty$, hence there is $R > 0$ such that

$$|P(z)| \geqslant \frac{1}{2}|z|^n \qquad \text{whenever } |z| > R.$$

Moreover, $|P|$ is bounded in $D(0, R)$ by the extreme value theorem. It follows from 2.15 that $1/P$ is constant, contradicting our assumption. Hence P has a root, say $z_0 \in \mathbb{C}$. We can write $P(z) = (z - z_0)Q(z)$ where $\deg Q < n$. Our claim follows. □

2.3. Further Theorems

Theorem 2.17. *Suppose Ω is a region and $f \in H(\Omega)$. Let $Z(f)$ denote the zero set of f. Then either $Z(f) = \Omega$ or $Z(f)$ has no limit points in Ω.*

We recall that a region is a nonempty open connected set. By limit points of $Z(f)$ we mean the set of all $z \in \mathbb{C}$ such that any neighbourhood of z intersects $Z(f)$.

Proof. Let A denote the set of limit points of $Z(f)$. Since f is continuous, $A \subset Z(f)$.

Fix any $a \in Z(f)$ and pick $r > 0$ such that $D(a, r) \subset \Omega$. By 2.13,

$$f(z) = \sum_{n=0}^{\infty} c_n (z - a)^n. \qquad (z \in D(a, r))$$

We now have two cases, either all $c_n = 0$ in which case $D(z_0, r) \subset A$ and a is an interior point of A; or there is a (strictly positive) smallest integer m for which $c_m \neq 0$. In this case, define

$$g(z) = \begin{cases} (z-a)^{-m} f(z) & z \in \Omega \setminus \{a\} \\ c_m & \text{otherwise} \end{cases}.$$

Then $g \in \mathcal{H}(\Omega \setminus \{a\})$ and

$$g(z) = \sum_{k=0}^{\infty} c_{m+k}(z-a)^k. \qquad (z \in D(a,r))$$

The above expansion implies that $g \in \mathcal{H}(\Omega)$. Since $g(a) \neq 0$, the continuity of g gives a small neighbourhood of a where g is not zero, hence a is an isolated point of $Z(f)$ and hence $a \notin A$ in this case.

Suppose A is nonempty, then we have shown that each point of A is an interior point, i.e. A is open. Moreover the set of limit points of any set is also closed. Since Ω is connected and A is a nonempty clopen subset of Ω, we get that $A = \Omega$. Hence $Z(f) = \Omega$. □

Corollary 2.18. *Let $f, g \in H(\Omega)$ and if $f(z) = g(z)$ for all z on a set which has a limit point in Ω then $f \equiv g$.*

Theorem 2.19. *(Maximum Modulus Principle) Let Ω be a region and $f \in \mathcal{H}(\Omega)$. If there is $z_0 \in \Omega$ such that $|f|$ attains a local maximum at z_0, then f is constant.*

Proof. Let δ be such that $|f(z_0)| \geqslant |f(z)|$ for all $z \in \overline{B}(z_0, \delta)$. Let $\gamma(t) = z_0 + re^{it}$ for $t \in [0, 2\pi]$ and $r \leqslant \delta$. Using Cauchy's Theorem, we have

$$f(z_0) = \frac{1}{2\pi} \int_\gamma \frac{f(\zeta)}{\zeta - z_0} d\zeta = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{i\theta}) d\theta.$$

Now,

$$|f(z_0)| \leqslant \frac{1}{2\pi} \int_0^{2\pi} |f(z_0 + re^{i\theta})| d\theta \leqslant |f(z_0)|$$

Hence,

$$\int_0^{2\pi} (|f(z_0)| - |f(z_0 + re^{i\theta})|) d\theta = 0$$

Since the integrand is positive, we get that $|f(z_0)| = |f(z_0 + re^{i\theta})|$ for all $\theta \in [0, 2\pi]$. Since r was arbitrary, we get that f is constant in $\overline{B}(z_0, \delta)$ (Hint: using the Cauchy-Reimann equations show that $\mathrm{Re}(f)$ and $\mathrm{Im}(f)$ are constant). It follows from 2.18 that f is constant in Ω. □

The above theorem has a useful corollary. If $f \in \mathcal{H}(\Omega)$ with compact closure $\overline{\Omega}$ and $f \in C(\overline{\Omega})$, then

$$\sup_{z \in \Omega} |f(z)| \leqslant \sup_{z \in \partial\Omega} |f(z)|.$$

This is a straight application of the maximum value theorem.

Remark 2.20. Suppose $f_j \in \mathcal{H}(\Omega)$ for all $j \in \mathbb{N}$ and $f_j \rightrightarrows f$ on compact subsets of Ω. Then $f \in H(\Omega)$. Further $f_j' \rightrightarrows f'$ on compact subsets of Ω.

Proof. To show that $f \in \mathcal{H}(\Omega)$, we use 2.14. Let Δ be a closed triangle in Ω, note that Δ is compact. Using 2.6, we get

$$\int_{\partial\Delta} f(z) dz = \lim_{n \to \infty} \int_{\partial\Delta} f_n(z) dz = 0.$$

To show the second part of the proposition, let $K \subset \Omega$ be compact.

For each $z \in K$, there is $r_z > 0$ such that $D(z, r_z) \subset \Omega$. Observe that $\{B(z, r_z)\}_{z \in K}$ is an open cover of K, and by compactness, there is $r > 0$ such that $D(z, r) \subset \Omega$ for all $z \in K$.

(TODO)

Applying 2.12 to $f - f_n$ on E, we get

$$|f'(z) - f_n'(z)| \leqslant r^{-1} \| f - f_n\|_E$$

where $\| f - f_n\|_E$ denotes the supremum of $f - f_n$ on E. Since $f_n \rightrightarrows f$ on E, we get that $f_n' \rightrightarrows f'$ on E, and hence on all compact subsets of Ω. □

Lemma 2.21. *If $f \in \mathcal{H}(\Omega)$ and g is defined in $\Omega \times \Omega$ as*

$$g(z,w) = \begin{cases} \dfrac{f(z) - f(w)}{z - w} & z \neq w \\ f'(z) & \textit{otherwise} \end{cases}$$

then g is continuous in $\Omega \times \Omega$.

Proof. It suffices to show continuity at the points $(a,a) \in \Omega \times \Omega$. Fix $a \in \Omega$ and $\epsilon > 0$. Then there is $r > 0$ such that $B(a,r) \subset \Omega$ and $|f'(\zeta) - f'(a)| < \epsilon$ for all $\zeta \in B(a,r)$. For $z, w \in B(a,r)$, define the path

$$\zeta(t) = (1-t)z + tw. \qquad (t \in [0,1])$$

Now,

$$g(z,w) - g(a,a) = \int_0^1 (f'(\zeta(t)) - f'(a))dt.$$

The absolute value of the integrand is $< \epsilon$, hence $|g(z,w) - g(a,a)| < \epsilon$. This proves our assertion. □

Remark 2.22. (Open Mapping Theorem) If Ω is a region and $f \in \mathcal{H}(\Omega)$ then $f(\Omega)$ is either a region or a point.

Remark 2.23. (Inverse Function Theorem) Suppose $\varphi \in \mathcal{H}(\Omega)$, $z_0 \in \Omega$ and $\varphi'(z_0) \neq 0$. Then Ω contains a neighbourhood V of z_0 such that

(a) φ is injective in V,

(b) $W = \varphi(V)$ is open,

(c) if $\psi : W \to V$ is defined by $\psi(\varphi(z)) = z$, then $\psi \in \mathcal{H}(W)$.

2.4. Homotopy

Remark 2.24.

2.5. Calculus of Residues

Remark 2.25. If $a \in \Omega$ and $f \in \mathcal{H}(\Omega \setminus \{a\})$, then f has a **removable singularity** at a if f can be so defined at a such that the extended function is holomorphic on Ω.

Remark 2.26. Suppose $f \in \mathcal{H}(\Omega \setminus \{a\})$ and f is bounded in neighbourhood around a, then f has a removable singularity at a.

Remark 2.27. If $a \in \Omega$ and $f \in \mathcal{H}(\Omega \setminus \{a\})$, then one of the following occurs:

(a) f has a removable singularity at a,

(b) There are $c_1, \dots, c_n \in \mathbb{C}$ such that

$$f(z) - \sum_{k=1}^{n} \frac{c_n}{(z-a)^k}$$

has a removable singularity at a.

(c) If $r > 0$ and $D(a,r) \subset \Omega$, $f(D'(a,r))$ is dense in $\mathbb{C}$.

In case (b), f is said to have a *pole* of order n at a. Further, the function

$$\sum_{k=1}^{n} c_n(z-a)^{-k}$$

is called the principal part of f at a. In case (c), f is said to have an *essential singularity* at a.

Remark 2.28. A function f is said to be *meromorphic* in an open set Ω if there is a set $A \subset \Omega$ such that

(a) A has no limit point in Ω

(b) $f \in \mathcal{H}(\Omega \setminus A)$

(c) f has a pole at each $a \in A$.

Observe that A is at most countable since a compact set must contain only finitely many point of A.

Remark 2.29. Suppose f is a meromorphic function in Ω and let $A \subset \Omega$ be the set of poles of Ω.

Remark 2.30. (Rouche's Theorem) Suppose that $f, g \in \mathcal{H}(\Omega)$. Let $D(a,r) \subset \Omega$ and let $C = \partial D$. If

$$|f(z) - g(z)| < |f(z)| \qquad \forall z \in C,$$

then f and g have the same number of zeros (counted with multiplicity) in $B(a,r)$.

Remark 2.31. LOG stuff

3. Unsorted

3.1. The Gamma Function

Remark 3.1. For $\mathrm{Re}(s) > 0$, define the Gamma function

$$\Gamma(s) = \int_0^\infty e^{-t} t^{s-1} dt \tag{6}$$

We prove that this formula is infact meaningful.

$$\Gamma(s) = \int_0^1 e^{-t} t^{s-1} dt + \int_1^\infty e^{-t} t^{s-1} dt$$

We first look at the first integral. The integrand is smaller than t^{s-1} when $t > 0$, hence

$$\int_0^1 e^{-t} t^{s-1} dt < \lim_{\epsilon \to 0} \int_\epsilon^1 t^{s-1} dt = \lim_{\epsilon \to 0} \left(\frac{1}{s} - \frac{\epsilon^s}{s} \right) = \frac{1}{s}.$$

Now, we look at the second integral. When $t > 0$, then every term of the expansion of e^t is positive, hence the inequality $e^t > t^n/n!$ holds for all positive integers n. Choosing $n > s+1$, we get

$$\int_1^\infty e^{-t} t^{s-1} dt = \lim_{\delta \to \infty} \int_1^\delta e^{-t} t^{s-1} dt < n! \lim_{\delta \to \infty} \int_1^\delta t^{s-n-1} dt = \frac{n!}{n-s}.$$

Remark 3.2. We can extend the gamma function to $\operatorname{Re} z > 0$ by again defining it as in 6. We use the fact that for $a, z \in \mathbb{C}$,

$$\operatorname{Re} a^z = \operatorname{Re} e^{z \log a} = e^{\operatorname{Re} z \log a} = a^{\operatorname{Re} z}.$$

Remark 3.3. Using integration by parts,

$$\Gamma(s+1) = \int_0^\infty e^{-t} t^s ds = -e^{-t} t^s \Big|_0^\infty + s \int_0^\infty e^{-t} t^{s-1} dt = s\Gamma(s).$$

Hence, for nonnegative integers we have

$$\Gamma(n+1) = n! \qquad n = 0, 1, 2, \ldots$$

https://ncatlab.org/nlab/files/Artin-TheGammaFunction.pdf

3.2. Misc

Remark 3.4. (Holomorphic functions defined by integrals) Let $\Omega \subset \mathbb{C}$ be open and let $K : [a, b] \times \Omega \to \mathbb{C}$ be such that

1. K is continuous on $[a, b] \times \Omega$,
2. For a fixed $t \in [a, b]$, $K(t, -) \in \mathcal{H}(\Omega)$,
3. $\partial_z K(t, z)$ is continuous in t.

then $f : \Omega \to \mathbb{C}$ defined as

$$f(z) = \int_a^b K(t, z) dt$$

is holomorphic on Ω.

Proof. https://people.math.osu.edu/gautam.42/S20/LectureNotes/Lecture30.pdf ☐

References

[Ahl78] Lars V Ahlfors. *Complex Analysis*. International series in pure and applied mathematics. McGraw-Hill Professional, New York, NY, 3 edition, December 1978.

[Con95] John B Conway. *Functions of one complex variable*. Graduate Texts in Mathematics. Springer, Berlin, Germany, 1995.

[Lan10] Serge Lang. *Complex Analysis*. Graduate Texts in Mathematics. Springer, New York, NY, October 2010.

[Rud76] Walter Rudin. *Principles of mathematical analysis*. International series in pure and applied mathematics. McGraw-Hill Professional, New York, NY, 3 edition, February 1976.

[Rud86] Walter Rudin. *Real and Complex Analysis*. McGraw-Hill series in higher mathematics. McGraw-Hill Professional, New York, NY, 3 edition, September 1986.

[SS03] Elias M Stein and Rami Shakarchi. *Complex analysis*. Princeton lectures in analysis. Princeton University Press, Princeton, NJ, April 2003.